

Flex

KC Sivaramakrishnan

IIT Madras

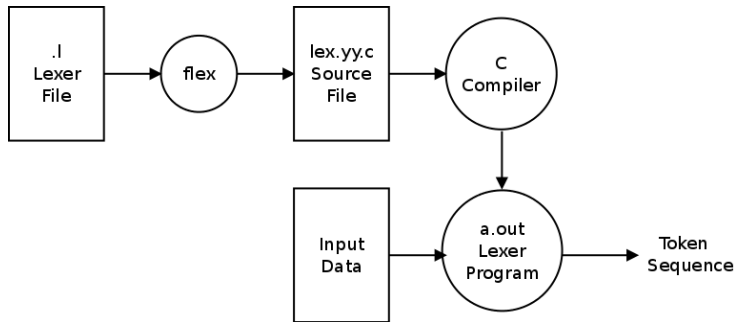
Monsoon 2026

Administrativa

- 6 assignments, **15 marks each** (90 marks, 40% of the course grade)
 - **9 marks** for the take-home part.
 - **6 marks** for an extension you write during the lab session after the deadline, on DCF machines, with no network access other than Moodle.
- Assignments are done individually, take-home and lab alike
 - **No LLM use is permitted**, for any part of any assignment.
 - MOSS runs on every submission, including against previous years.
- **Attendance at the six in-lab evaluation sessions is mandatory**
 - That is where the 6 marks are decided, so a missed session cannot be made up later.
 - Miss one without an institute-approved reason and your take-home marks for that assignment are capped at 20%.

Flex

- Flex stands for *Fast Lexical Analyzer generator*
- Tool for generating lexical analyzers
- Manual: <https://www.cs.virginia.edu/~cr4bd/flex-manual/index.html>



Compiling a lex program

```
$ flex demo1.l  
$ gcc -o demo1.exe lex.yy.c -lfl  
$ ./demo1.exe
```

Format of the input file

definitions

%%

rules

%%

user code

Definitions section

Declarations of simple name definitions to simplify the scanner specification.

They are of the form `name definition` *unintended*.

For example,

- `DIGIT [0-9]`
- `ID [a-z][a-z0-9]*`

Rules section

Defines RE patterns and actions to take when those patterns are encountered.

They are of the form `pattern action` *unintended*.

Actions can be arbitrary C statements. If the action is empty, then the input text that matched the pattern are discarded.

Patterns are regular expressions. See <https://www.cs.virginia.edu/~cr4bd/flex-manual/Patterns.html>.

User code section

Any code copied verbatim to the output.

Values available to the user

```
char *yytext
```

holds the text of the current token. NULL terminated. It may be modified but not lengthened (you cannot append characters to the end).

```
int yyleng
```

holds the length of the current token.

Demos